

Conservation of energy (TSec. 2.7)

Statement of conservation of energy - first law of thermodynamics
i.e., energy in an isolated system cannot be created or destroyed (it is constant) it can only be transformed from one form to another.
For a **closed system**,

$$\begin{aligned} \{\text{change in energy}\} &= \{\text{work done on the system}\} \\ &+ \{\text{heat transfer to the system}\} \end{aligned} \quad (1.21)$$

$$\rightarrow dE = dW + dQ \quad (1.22)$$

where, E is energy, W is work done and Q is heat transfer. Your textbook (TEq. (2.2)) uses lower case e , w and q . This equation (TEq. (2.2)) in your textbook is in terms of per unit mass, hence the use of lowercase letters.

Cont. : changes in mass in CV + transport of mass through surface
= ,

Mom : change in Mom. in CV + " " mom. " "
= Force

Conservation of energy

Energy: change in Energy in C.V + " " Energy " "

= W + Q
↓ ↓
work heat
transfer

Considering an open system,

{rate of change in total energy} =

- {net rate of change of total energy flow out of (vol) by convection}

+ {net rate of work done on (vol) by the surroundings}

+ {net rate of heat added to (vol) other than by convection}

(1.23)

$$E_t = \underbrace{\int e}_{\text{internal energy}} + \underbrace{\frac{1}{2} \int \mathbf{v} \cdot \mathbf{v}}_{\text{k.E}} + \int g z$$

Total energy

Total energy = internal energy + kinetic energy + potential energy

$$E_t = \underbrace{\int \rho u_e}_{\text{internal energy}} + \underbrace{\frac{1}{2} \int \rho [u^2 + v^2 + w^2]}_{\text{kinetic energy}} + \underbrace{\int \rho g z}_{\text{potential energy}} \quad (1.24)$$

$$E_t = \rho e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + \rho g z \quad (1.25)$$

where, u_e is the internal energy per unit mass, u, v & w are the x, y & z components of $\vec{V}$, g is the acceleration due to gravity and z the ordinate in the direction in which the potential energy is changing (height).

Total energy

Therefore,

$$\{ \text{rate of change in } E_t \} = \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) d(\text{vol}) \quad (1.26)$$

u_e
 $e'' = \sqrt{V}$

{rate of change of E_t flow out of (vol) by convection} =

$$\oiint_A \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) \vec{V} \cdot \hat{n} dA \quad (1.27)$$



mass going out/in from C.S

mass $\times E$ going out from surface

$$\text{Energy Eq: } \frac{\partial}{\partial t} \int_{\text{Vol}} m E + \int \vec{V} \cdot d\vec{s} = W + Q$$

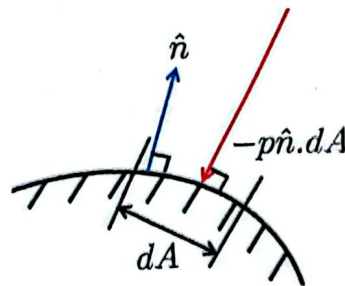
Work done by forces

$$W = F \cdot dx$$

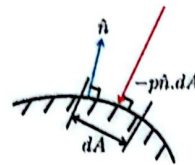
$$\text{rate of } W = \dot{W} = F \cdot \frac{dx}{dt}$$

$$\text{work} = \text{force} \times \text{distance} \quad (1.28)$$

$$\text{rate of work} = \text{force} \cdot \frac{\text{distance}}{\text{time}} = \text{force} \cdot \text{velocity} = \underline{\underline{\vec{F} \cdot \vec{V}}} \quad (1.29)$$



Work done by forces



$$-\oint (p \vec{n} dA) \cdot \vec{V} : \text{pressure work}$$

$$\{\text{net rate of } W \text{ done by } p\} = - \oint_A p \vec{V} \cdot \vec{n} dA \quad (1.30)$$

$$\{\text{net rate of } W \text{ done by body forces e.g., gravity}\} = \iiint_{\text{vol}} \rho \vec{f}_b \cdot \vec{V} d(\text{vol}) \quad (1.31)$$

$$\frac{f_b}{\text{mass}} : \text{body force}$$

$$\{\text{net rate of } W \text{ done by viscous forces}\} = \dot{W}_{\text{visc}} \quad (1.32)$$

$$\{\text{Shaft work}\} = \dot{W}_s \quad (1.33)$$

Here shaft work is work done through a shaft e.g., turbine, pump

Conservation of energy

As before, **global or the integral form** of the conservation of energy becomes,

$$\begin{aligned} & \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) d(\text{vol}) = \text{change of energy in c.v.} \\ & - \iint_A \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) \vec{V} \cdot \hat{n} dA \text{ transport of energy through surfaces} \\ & + \dot{Q} - \underbrace{\iint_A p \vec{V} \cdot \hat{n} dA}_{\text{H.T}} + \underbrace{\dot{W}_{\text{visc}} + \dot{W}_s}_{\text{P-Work}} + \iiint_{\text{vol}} \rho \vec{f}_b \cdot \vec{V} d(\text{vol}) \quad (1.35) \end{aligned}$$

Note, the equivalent equation in your textbook (TEq. (2.37)) is in a slightly different form. The differences are highlighted in the following slide.

Conservation of energy

Using the same procedure as before (divergence theorem etc.) the **local or differential form** of the conservation of energy becomes.

$$\frac{\partial}{\partial t} \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) + \nabla \cdot \left[\rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) \vec{V} \right] = \dot{q} - \nabla \cdot (p \vec{V}) + \dot{w}_{visc} + \rho \vec{f}_b \cdot \vec{V} \quad (1.36)$$

Differential form
of energy Eq.

If steady, inviscid, w.o body force, adiabatic ($\dot{q}=0$)
 $\frac{\partial}{\partial t} = 0$ $\dot{w}_{vis} = 0$ $\vec{f}_b = 0$

AE 319, EPG Cons. Eqns., Cons. Energy, 33

$$\Rightarrow \nabla \cdot \left[\rho \left(\overset{\text{internal energy}}{u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz} \right) \vec{V} \right] + \nabla \cdot (p \vec{V}) = 0$$

$$\nabla \cdot \left[\rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz + \frac{p}{\rho} \right) \vec{V} \right] = 0$$

Assume 1-D, ignore gz (i.e., gas) $\Rightarrow \frac{\partial}{\partial x} \left(\rho \left(u_e + \frac{1}{2} V^2 + \frac{p}{\rho} \right) \right) = 0$

$$\frac{\partial}{\partial x} = 0 \Rightarrow \rho \left(u_e + \frac{1}{2} V^2 + \frac{P}{\rho} \right) = \text{cte}$$

$$\text{Enthalpy: } h = u_e + \frac{P}{\rho} \Rightarrow h + \frac{1}{2} V^2 = \text{cte}$$

one-Dimensional flow

$$V = u_x$$

$$\rightarrow h + \frac{u_x^2}{2} = \text{cte}$$

$$h = c_p T$$

$$c_p T + \frac{u_x^2}{2} = \text{cte}$$

integrated energy Eq.

Inviscid Flow (TSec. 3.1)

Focus for a while on the differential form of the momentum eq. (1.20)

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} + \frac{1}{\rho} (\vec{F}_{visc} + \vec{F}_{ext})$$

Assume the flow is inviscid (negligible viscous forces), and no other external forces.

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} \quad (\text{TEq. (3.2)}) \quad (2.5)$$

↓

body force

When is the flow inviscid?

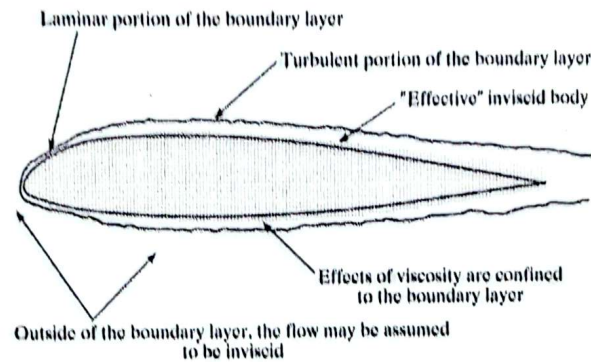


Figure 2.1: Fig. 2.14 from AE-BC

- ▶ Outside the boundary layer and wake - potential flow (inviscid, irrotational & steady - more on this soon)
- ▶ Boundary layer and wake - unsteady, viscous & rotational

NOTE: Remember inviscid only means that viscous forces are negligible compared to other forces in the system. If viscosity is zero then it is not a fluid.

When is the flow inviscid?

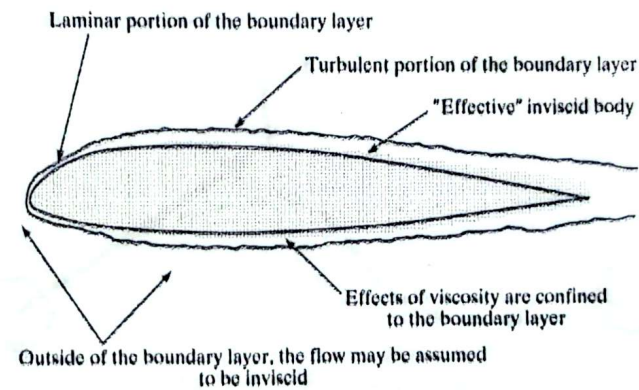


Figure 2.2: Fig. 2.14 from AE-BC

Note,

$$\text{Reynolds number} = \frac{\text{Inertial forces}}{\text{Viscous forces}} \quad (2.6)$$

Therefore, in the case of inviscid flow,

$$\text{Reynolds number} \rightarrow \infty \quad (2.7)$$

Pathlines - AE 201 revision

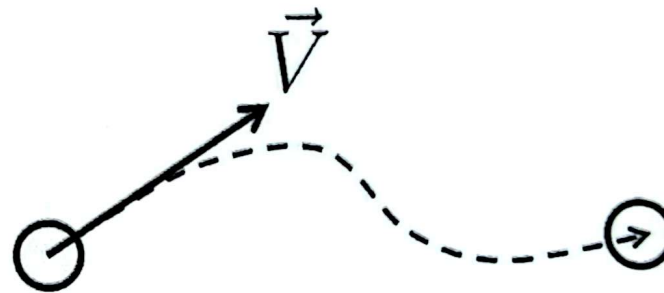


Figure 2.3: A pathline followed by a fluid element. Also shown is the velocity vector at the initial starting point.

Pathlines are paths followed by individual fluid elements. We usually obtain this by integrating in time. We start at a point and integrate forward.

Streamlines - AE 201 revision

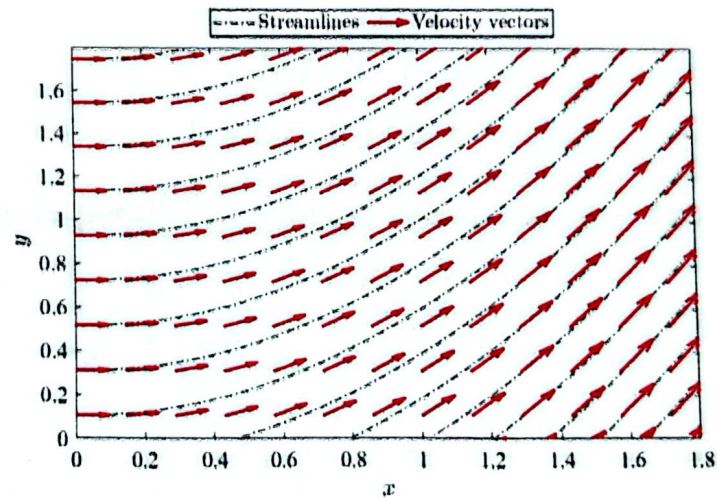
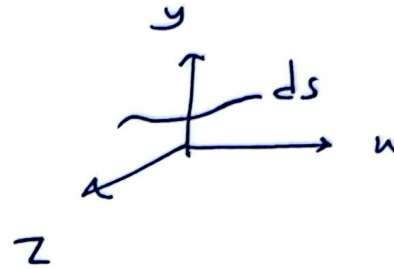


Figure 2.4: Streamlines in a flow. The black lines are the streamlines while the red vectors are the velocity vectors which are tangent everywhere to the streamlines.

Streamline: - is a curve whose tangent at every point is the direction of the velocity vector $\vec{V}$ at that point



Streamlines - AE 201 revision

Let $\vec{ds}$ be the streamline in a flow and let $\vec{V}$ be the velocity vector.

$$\vec{ds} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\vec{V} = u \hat{i} + v \hat{j} + w \hat{k}$$

$\vec{V}$ is tangent to $\vec{ds}$
so their cross product
is zero

$$\vec{ds} \times \vec{V} = ds \vec{V} \sin \theta$$

$$\theta = 0 \rightarrow \sin \theta = 0$$

$$\Rightarrow \vec{ds} \times \vec{V} = 0$$

For a vector to be parallel to another vector the curl has to be zero.

$$\vec{ds} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ dx & dy & dz \\ u & v & w \end{vmatrix} = 0 \quad (2.8)$$

$$= \underbrace{(w dy - v dz)}_A \hat{i} + \underbrace{(u dz - w dx)}_B \hat{j} + \underbrace{(v dx - u dy)}_C \hat{k} = 0 \quad (2.9)$$

Therefore, the equation of a streamline:

$$w dy - v dz = u dz - w dx = v dx - u dy = 0 \quad (2.10)$$

like (2.9)
when a vector is zero
individual components
should be zero

So $A=B=C=0$

which gives us "2.10" $\rightarrow$ "stream line Equation"

Angular velocity is defined as $(\vec{\omega})$ $\frac{1}{2} \vec{\nabla} \times \vec{V} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$

Irrotational & rotational flow (TSec. 3.6)

In 3-D the angular velocity:

$$\vec{\omega} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \frac{1}{2} \vec{\nabla} \times \vec{V}$$

In cylindrical coordinates the angular velocity:

cyl. coord. $\rightarrow$
$$\vec{\omega} = \frac{1}{2r} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ v_r & rv_\theta & v_z \end{vmatrix} = \frac{1}{2} \vec{\nabla} \times \vec{V}$$

Note, here $\times$ is the "curl" operator. Also, $\vec{\omega}$ is a vector.

"Cartesian coordinate"

In 2D, $\vec{\nabla} \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

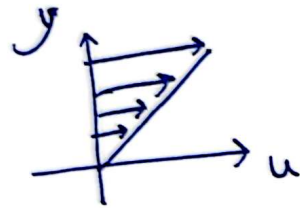
So Ex. in HW #1, which

(2.11) has $\begin{cases} u = u(y) \\ v = v(x) \end{cases}$

is rotational flow

(2.12)

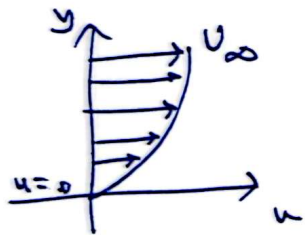
Ex. if $u = u(y)$, this flow is rotational! $\omega_z = \frac{\nabla \times \vec{V}}{2} = \frac{1}{2} \frac{\partial u}{\partial y}$



Irrotational and rotational flow

A fluid element in an irrotational flow only translates and deforms. It does not rotate. Mathematically, a flow is irrotational when

$$\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0 \quad (\text{TEq. (3.17)}) \quad (2.13)$$



$\frac{\partial u}{\partial y} \neq 0 \rightarrow$ Flow in boundary layer is rotational

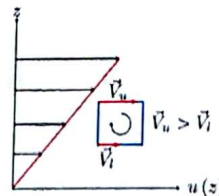


Figure 2.5: Fluid element in a flow field rotating due to the velocity gradient it is subject to

Viscous flows ARE rotational. Some inviscid flows are rotational.

Irrotational and rotational flow

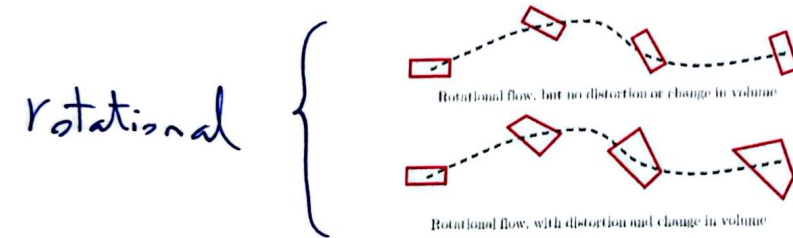


Figure 2.6: Rotational flow

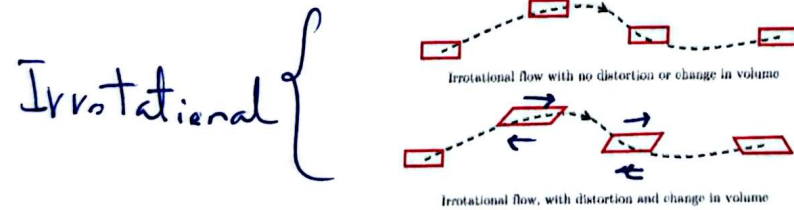


Figure 2.7: Irrotational flow

Net rotation is zero!

Circulation (TSec. 3.5)

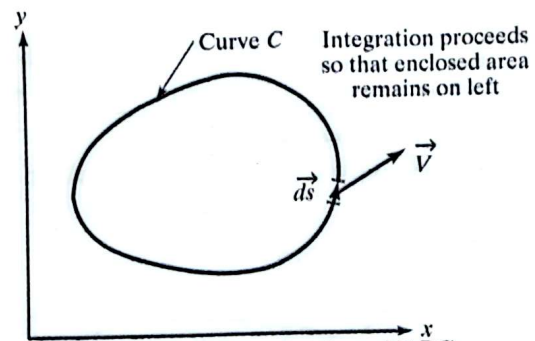


Figure 2.8: Fig. 3.4 from AE-BC

$$\text{Circulation } \Gamma = - \oint_C \vec{V} \cdot d\vec{s} \quad (\text{TEq. (3.14)}) \quad (2.14)$$

The $\oint_c$ here indicates integral over a closed surface. Also, circulation Γ here is defined as being positive in a clockwise sense (Ensures positive lift!)

$$\Gamma = - \oint \underbrace{V \cdot ds}_{\text{length}} = \iint \underbrace{(\nabla \times \vec{V}) \cdot d\vec{s}}_{\text{area}} = - \iint (\vec{\nabla} \times \vec{V}) \cdot \vec{n} dA$$

If a flow is irrotational ($\vec{\nabla} \times \vec{V} = 0$)

$$\Rightarrow \boxed{\Gamma = 0}$$

Circulation

Using Stokes theorem,

$$\Gamma = - \oint_C \vec{V} \cdot d\vec{s} = - \iint_A (\vec{\nabla} \times \vec{V}) \cdot \vec{n} dA \quad (\text{TEq. (3.16)}) \quad (2.15)$$

Therefore, if the flow within the contour of integration is irrotational everywhere, $\Gamma = 0$ as $\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0$

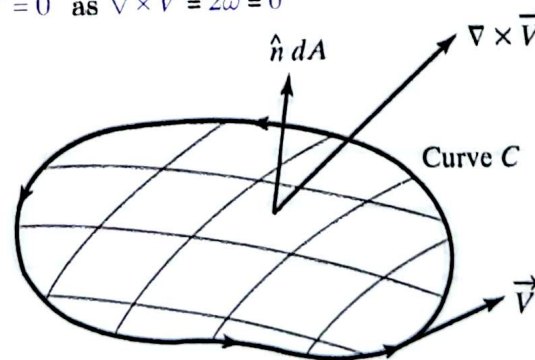
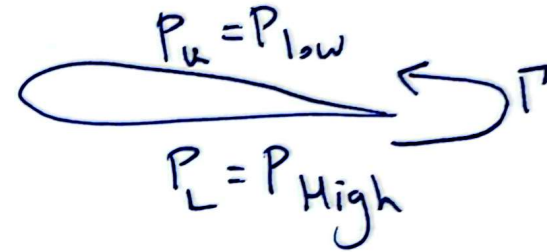


Figure 2.9: Fig. 3.6 from AE-BC



Circulation

Flow having circulation does not mean fluid particles are going around in circles! Typically the streamlines need to have some sort of curvature for finite circulation to exist. In other words the flow has to turn in some manner as in flow over an airfoil

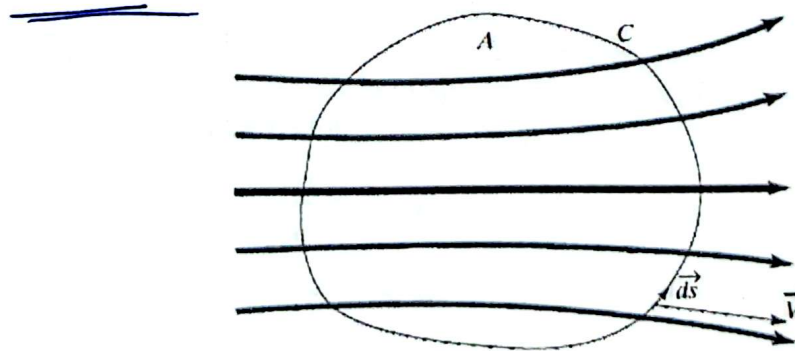


Figure 2.10: from AE-BC

stagnation density ρ_t $\leftarrow \rho_t = \left(1 + \frac{\gamma-1}{2} M_\infty^2\right)^{\frac{1}{\gamma-1}}$

static density $\rho \leftarrow \rho$

If $M = 0.3$

$\frac{\rho_t}{\rho} = 1.0456 \rightarrow \Rightarrow \rho = \text{cte} \quad \boxed{\text{incomp.}}$

stagnation

When is flow incompressible?

In general, compressibility effects are insignificant at low speeds and the density can be assumed to be constant.

$$\frac{\rho_{t1}}{\rho} = \left(1 + \frac{\gamma-1}{2} M_\infty^2\right)^{1/(\gamma-1)} \quad (\text{TEq. (8.36)}) \quad (2.16)$$

Eq. (2.16) will be derived later in the course. Here, ρ_{t1} is the stagnation density and γ is the ratio of specific heats (see (TEq. (8,15))).

Note for air ($\gamma = 1.4$) at $M_\infty = 0.3$

$$\frac{\rho_{t1}}{\rho} = 1.0456 \quad \text{i.e., 5\% variation in } \rho \quad (2.17)$$

Therefore, as a rule of thumb, we assume the flow to be incompressible when $M_\infty \leq 0.3$.

Note, nothing magical happens between $M_\infty = 0.29$ and $M_\infty = 0.31$.

we want to solve momentum Eq. ~~by~~ in a simplified form $\rightarrow$ "potential flow"

Hw. 1) Is this flow physical? check $\nabla \cdot \vec{V} = 0$.

Hw. 3) Is " " irrotational? check $\nabla \times \vec{V} = 0$.

Potential flow assumptions

Under certain conditions, parts of the flow can be assumed to be

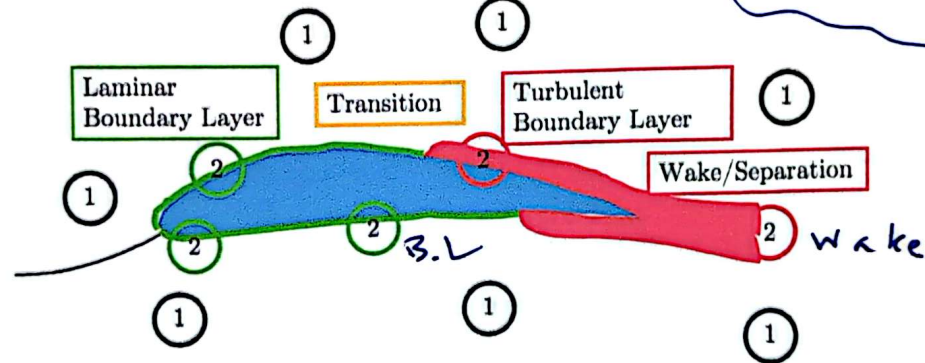
1. Steady i.e., $\frac{\partial(\quad)}{\partial t} \approx 0$
2. Incompressible i.e., $M_\infty \leq 0.3$ or $\rho \approx \text{constant}$
3. Irrotational (implies inviscid) i.e., $\nabla \times \vec{V} = 0$

Collectively, when these assumptions are imposed on a flow field we consider the flow to be "ideal" or "potential" flow. I call these the 3.5 assumptions of potential flows.

Initially we shall focus on 2-D potential flows (airfoils) and then extend it to 3-D flows (wings). In aerodynamics, when we consider flow over airfoils, wings etc., everything is at the same altitude so we neglect gravitational forces (the most common form of body force). But this is not an ideal or potential flow assumption. So to be rigorous we are going to consider potential flows with no body force effects.

can predict lift
"cannot" $\nearrow$ "drag"

Potential flow



- ▶ Region 1 - Outside the boundary layer and wake at low speeds the flow can be considered to be potential.
- ▶ Region 2 - Within the boundary layer and wake, even at low incompressible speeds, the flow is rotational (~~inviscid~~) and unsteady. viscous

viscous

viscous

② { B.L
wake

flow is
rotational
viscous
unsteady

Potential function (TSec. 3.8)

2-D mass conservation (Eq. (1.12)):

$$\frac{\partial \rho}{\partial t} + \bar{\nabla} \cdot (\rho \bar{V}) = 0 \quad (2.18)$$

$$\text{Steady: } \rightarrow \frac{\partial \rho}{\partial t} \approx 0 \quad (2.19)$$

$$\text{Incompressible: } \rightarrow \underline{\underline{\rho \approx \text{constant}}} \quad (2.20)$$

Therefore, 2-D continuity equation for steady, incompressible flow:

$$\bar{\nabla} \cdot \bar{V} = 0 \rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (2.21)$$

Potential fn. $\rightarrow$ irrotational flow $\vec{\nabla} \times \vec{V} = 0$ ①

from math

Consider an irrotational flow, then:

$$\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0$$

$$(2.22) \quad \vec{\nabla} \times (\vec{\nabla} \phi) = 0 \quad (2)$$

Vector calculus relationship:

$$\vec{\nabla} \times (\vec{\nabla} \phi) = 0, \text{ where } \phi \text{ is a scalar} \quad (2.23)$$

$$\text{Therefore, } \vec{V} = \vec{\nabla} \phi \quad (2.24)$$

curl of gradient
of any function is
zero!

That is IF the flow is irrotational then, the velocity vector $\vec{V}$ can be expressed as the gradient of a scalar ϕ . This scalar is called the velocity potential function or simply the velocity potential.

① and ② gives:

$$\boxed{\vec{V} = \vec{\nabla} \phi}$$

Velocity of the irrotational flow is gradient of a scalar ϕ , ϕ is called potential function!

In HW 3) find potential function if u & v are given!

$$\vec{V} = \vec{\nabla} \phi$$

$$\left\{ \begin{array}{l} u = \frac{\partial \phi}{\partial x} \\ v = \frac{\partial \phi}{\partial y} \\ w = \frac{\partial \phi}{\partial z} \end{array} \right.$$

Potential fn.

$$\text{Therefore, } \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \quad (\text{TEq. (3.19)}) \quad (2.25)$$

$$\text{Therefore, } u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}, \quad w = \frac{\partial \phi}{\partial z} \quad (2.26)$$

Chain rule (2-D):

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy \quad (2.27)$$

$$= u dx + v dy \quad (2.28)$$

Lines of equal potential or equipotential lines: $\rightarrow d\phi = 0$ Therefore, one way to describe equipotential lines is

$$\left. \frac{dy}{dx} \right|_{\phi=c} = -\frac{u}{v} \quad (2.29)$$

In 2D

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$d\phi = u dx + v dy = 0$$

$$\Rightarrow \left(\frac{dy}{dx} = -\frac{u}{v} \right) \text{ over equipotential lines}$$

2-D Cylindrical coordinates

$$\text{Velocities: } v_r(r, \theta) \quad v_\theta(r, \theta) \quad v_z = 0 \quad (2.30)$$

$$\text{Potential function: } v_r = \frac{\partial \phi}{\partial r} \quad v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} \quad (2.31)$$

$$V_r = \frac{\partial \phi}{\partial r} \quad V_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta}$$

stream function (2D):

streamline: line parallel to velocity,
so $\frac{dy}{dx} = \frac{v}{u}$

Stream function (TSec. 3.9)

Eq. of a streamline from eq. (2.10) for 2-d flow is given by,

$$\frac{dy}{dx} = \frac{v}{u} \rightarrow u \, dy - v \, dx = 0 = d\psi \quad (2.32)$$

$$\text{Therefore, } d\psi = 0 \rightarrow \psi = \text{constant} \quad (2.33)$$

Therefore, $\psi(x, y) = c$ is a streamline where ψ is called the stream function and c is any constant. Note, two important facts:

- ▶ The flow field doesn't have to be potential to define a stream function.
- ▶ The velocity $\vec{V}$ is tangent at every point on the line $\psi = c$ hence if we consider two streamlines $\psi = c_1$ and $\psi = c_2$ the mass flow between these two lines should be conserved.

Stream function

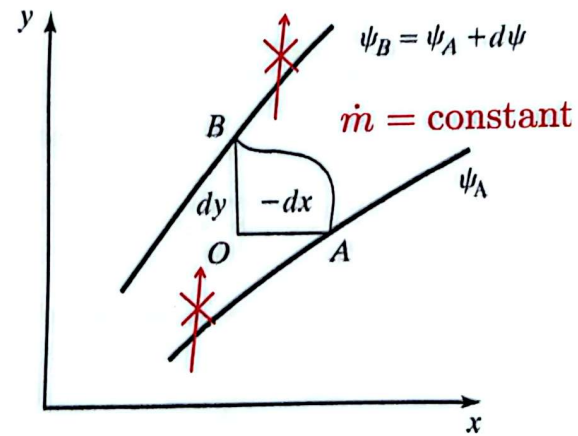


Figure 2.11: Modified Fig 3.7 from AE-BC

Between two streamline
 $\dot{m}$ is constant as the
 flow can not pass
 across the streamlines!

over streamline, $\frac{dy}{dx} = \frac{v}{u}$ (1)

$$d\psi = \frac{\partial\psi}{\partial x} dx + \frac{\partial\psi}{\partial y} dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\frac{\partial\psi}{\partial x}}{\frac{\partial\psi}{\partial y}}$$

Stream function

Use Chain rule:

$$d\psi = \frac{\partial\psi}{\partial x} dx + \frac{\partial\psi}{\partial y} dy$$

Therefore, from eq. (2.32)

$$u = \frac{\partial\psi}{\partial y}, v = -\frac{\partial\psi}{\partial x}$$

Hence the more formal way of writing the equation of a streamline is

$$\left. \frac{dy}{dx} \right|_{\psi=c} = \frac{v}{u}$$

(2)

Compare (1) and (2)

$$\Rightarrow \begin{cases} v = -\frac{\partial\psi}{\partial x} \\ u = \frac{\partial\psi}{\partial y} \end{cases}$$

(2.35)

(2.36)

So if I give you

u and v, you
can find ψ

which is streamfunction

Stream fn. for steady & incompressible flow

From eq. (2.35)

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

Therefore, $\nabla \cdot \vec{V} = 0$ (eq. (2.21)), which is the steady, incompressible continuity equation, becomes

$$\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} = 0$$

As long as ψ is a reasonable function then $\frac{\partial^2 \psi}{\partial x \partial y} = \frac{\partial^2 \psi}{\partial y \partial x}$.

Hence, for steady, incompressible flow ψ satisfies the continuity equation

$$\begin{aligned} \vec{\nabla} \cdot \vec{V} &= . \\ \frac{\partial}{\partial x}(u) + \frac{\partial}{\partial y}(v) &= . \\ \frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial x}\right) &= 0 \\ (2.37) \quad \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} &= . \checkmark \\ \text{So } \psi &\text{ satisfies} \\ &\text{the continuity Eq.} \end{aligned}$$

2-D Cylindrical coordinates

$$\text{Velocities: } v_r(r, \theta) \quad v_\theta(r, \theta) \quad v_z = 0 \quad (2.38)$$

$$\text{Stream function: } v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} \quad v_\theta = -\frac{\partial \psi}{\partial r} \quad (2.39)$$

cyl. coord relation of $\begin{cases} v_r \\ v_\theta \end{cases}$ with ψ

$$\phi = cte \rightarrow \frac{dy}{dx} = -\frac{u}{v} \quad \phi = cte$$

$$\psi = cte \rightarrow \frac{dy}{dx} = \frac{v}{u} \quad \psi = cte$$

Equipotential lines & streamlines (TSec. 3.10)

Lines of constant ϕ (equipotential lines)

$$d\phi = 0 \rightarrow u dx = -v dy \rightarrow \left. \frac{dy}{dx} \right|_{\phi=\text{const}} = -\frac{u}{v} \quad (2.40)$$

Lines of constant ψ (streamlines)

$$d\psi = 0 \rightarrow u dy = v dx \rightarrow \left. \frac{dy}{dx} \right|_{\psi=\text{const}} = \frac{v}{u} \quad (2.41)$$

Therefore,

$$\left. \frac{dy}{dx} \right|_{\phi=\text{const}} \times \left. \frac{dy}{dx} \right|_{\psi=\text{const}} = -1 \quad (2.42)$$

Therefore, equipotential lines are everywhere perpendicular to streamlines. Hence, if you know one you can calculate the other to within a constant.

$$\vec{\nabla} \cdot \vec{V} = 0 \xrightarrow{2D} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u = \frac{\partial \phi}{\partial x}$$

$$v = \frac{\partial \phi}{\partial y}$$

$$\text{put in } \vec{\nabla} \cdot \vec{V} = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$= \vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi = 0$$

Laplace eq. (TSec. 3.8 & 3.9)

Eq. (2.21) - Continuity equation for steady and incompressible flow:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 = \vec{\nabla} \cdot \vec{V}$$

Now if the flow is irrotational $\rightarrow \vec{V} = \vec{\nabla} \phi$. Substituting in eq. (2.21) we get,

$$\vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi = 0 = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \quad (2.43)$$

Further, as the flow is irrotational, $\vec{\nabla} \times \vec{V} = 2\vec{\omega} = 0$. Considering 2-D flow,

$$\vec{\nabla} \times \vec{V} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad (2.44)$$

$$\text{From eq. (2.35) } u = \frac{\partial \psi}{\partial y}, v = -\frac{\partial \psi}{\partial x}$$

So ϕ satisfies
Laplace Eq.